

electric potential - practice problems

name _____

1. A positive source charge creates electric potential at a nearby point. Is that potential positive, negative, or zero?
2. A $+2.0 \mu\text{C}$ charge and a $-3.0 \mu\text{C}$ charge are 0.20 m apart. What is the electric potential energy of the pair?
3. Two small metal spheres on insulating stands each have $+50 \text{ nC}$ of charge. How much work does it take to push them from 20 cm apart to 5.0 cm apart?
4. Two identical $+4.0 \mu\text{C}$ charges have 0.72 J of electric potential energy. How far apart are their centers?

5. What electric potential does a $+8.0 \text{ nC}$ source charge create at a point 12 cm away?

6. What electric potential does a -6.0 nC source charge create at a point 3.0 cm away?

7. A Van de Graaff generator's dome has a radius of 0.20 m . Its surface is at a potential of $200\,000 \text{ V}$. Treating the dome's charge as if it were all at the center, what charge is on the dome?

8. A $+3.0 \text{ }\mu\text{C}$ bead is at a location where the electric potential is -1200 V . What is the bead's electric potential energy there?

9. A $+2.0\ \mu\text{C}$ bead is placed $0.30\ \text{m}$ from a charged sphere. What is the bead's electric potential energy?

10. An electron is at a location where the electric potential is $+1500\ \text{V}$. What is the electron's electric potential energy?

11. A $+4.0\ \text{nC}$ charge is $20\ \text{cm}$ west of a point, and a $-2.0\ \text{nC}$ charge is $10\ \text{cm}$ east of the same point. What is the electric potential at the point?

12. A $+6.0\ \text{nC}$ charge is $30\ \text{cm}$ to the left of a point. A $-3.0\ \text{nC}$ charge is $10\ \text{cm}$ to the right of the point. What is the electric potential at the point?

13. A $+4.0 \text{ nC}$ charge is 20 cm west of a point, and a -2.0 nC charge is 10 cm east of the same point. The electric potential at the point is zero. What is the electric field at that point?

14. A $+2.0 \text{ }\mu\text{C}$ bead is moved from a location where the potential is 50 V to a location where the potential is 300 V. What is the change in its electric potential energy?

15. An electron starts from rest and is accelerated through a 1200 V potential difference. Using the nonrelativistic kinetic energy equation, what speed could it reach?

16. In Rutherford's 1911 experiment, alpha particles (charge $+2e$, mass $6.64 \times 10^{-27} \text{ kg}$) were fired at gold nuclei (charge $+79e$). An alpha particle moving at $1.5 \times 10^7 \text{ m/s}$ heads straight at a gold nucleus. Treating the gold nucleus as fixed in place, how close does the alpha particle get before it stops?